

Auditing Weather-Feature Reliance in Detrended Crop-Yield Models

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Abstract—Post-hoc explanations of a fitted predictive model should not be used as evidence about observed below-trend crop-yield events unless the same model has locked-period predictive adequacy for the same target. We audit this condition in a U.S. crop-state-year panel with 1257 rows for Barley, Canola, Oats, and Wheat from 1990–2025. Trends and residual scales are fit on training years only; selection uses 2012–2015 validation, and 2016–2025 remains locked until final evaluation. The validation-selected ExtraTrees weather only model has final $R^2 = -0.014$, RMSE 0.669 t ha^{-1} , and paired ΔRMSE versus a zero-residual baseline of $-0.005 [-0.019, 0.009]$. The selected model fails Module A, and Full-minus-Metadata Module B also fails with $\Delta\text{RMSE} -0.012 [-0.029, 0.002]$. Several pre-specified sensitivities are locally favorable, including the longer-history population, one rolling-origin fold, and severe-tail error components. They do not alter the primary decision because they evaluate a different population or fold, or fail the complete event-recovery rule. We therefore retain post-hoc explanation outputs only as descriptions of a fitted function. In synthetic stress tests, the observable A+B+E policy retains 88.9% sensitivity but only 28.7% specificity, showing that predictive modules alone do not reliably detect omitted confounding, leakage, or distributional violations. The contribution is a reproducible claim-eligibility protocol with paired year-block uncertainty, weather-value and event-recovery modules, calibration cases, and a warning that full-series detrending is retrospective only.

Index Terms—crop yield, explainable machine learning, temporal validation, negative results, reproducibility

I. INTRODUCTION

Machine-learning yield forecasts are increasingly built from weather, remote sensing, management, and crop-model variables. Studies report useful predictive systems across large and small panels, diverse model families, and hybrid formulations [1]–[4]. Yet a plausible feature attribution is not causal evidence, and agreement between explanation methods cannot compensate for a model that does not predict the future target reliably. This distinction is especially important for below-trend yield events, where a compelling explanation of an individual fitted prediction can be mistaken for an explanation of the observed loss.

Weather extremes and climate variability are materially associated with crop-yield variation at several spatial scales [5]–[9]. Those findings motivate careful predictive evaluation; they do not license an event-level attribution from this restricted panel. We ask a narrower question: does a leakage-safe residual model have enough future-period predictive adequacy and stability to permit substantive weather-feature-reliance interpretation? The answer is evaluated before XAI is interpreted.

The paper makes four contributions. First, it formulates a study-level claim-eligibility protocol for post-hoc explanations of predictive models. Second, it separates overall predictive adequacy (Module A), the conditional incremental value of weather beyond metadata (Module B), and below-trend event recovery (Module E); Weather-only versus Metadata-only remains a representation diagnostic (Module D). Third, it evaluates the rule in simulations with known mechanisms and pre-specified evaluation labels, revealing both abstention behavior and substantial false permission under structural invalidity, plus empirical abstention and permission cases. Fourth, it releases an audit trail that binds each module decision, interval, prediction vector, and target definition to a reproducible artifact. Train-only preprocessing and locked testing are protocol safeguards, not claimed as standalone novelty.

II. RELATED WORK

A. Yield Modeling and Detrending

Detrending separates a local historical yield trajectory from residual variation, but the result can depend on the detrending procedure and information available at the evaluation date [10], [11]. We therefore fit each crop-state trend and residual scale on training observations only. Ridge regression, Random Forest, and ExtraTrees provide deliberately distinct linear and tree-based model families [12]–[14]; the implementations use scikit-learn [15]. This is a finite pre-specified comparison, not a search over the locked test period.

B. Explanation Methods and Their Boundary

SHAP and LIME explain a model output under their respective approximation and reference choices [16], [17]. Permutation-style analyses, model reliance, accumulated local effects, and conditional importance likewise describe feature dependence or behavior of a fitted predictor [18]–[20]. They can be useful diagnostic artifacts. However, none turns failed prospective predictive adequacy into evidence about an observed yield anomaly. Prior work also shows that post-hoc explanations can be sensitive to perturbation behavior, model randomization, and deletion/retraining benchmarks, and that feature relevance has a causal interpretation problem unless the target estimand and reference distribution are explicit [21]–[25]. Those tests ask whether explanations are faithful to model behavior; the module policy below asks whether model behavior is eligible to support a locked-period outcome claim.

TABLE I
DATA-FLOW AND FINAL-TEST ELIGIBILITY RECONSTRUCTED FROM THE
RAW INPUTS.

Stage	Rows	Rule
raw yield file	1291	source table supplied in repository
analysis year filter	1257	exclude 34 raw 1989 rows because analysis starts in 1990
processed model frame	1257	exact reconstruction from raw yields plus NASA POWER daily input
final test eligible	333	2016-2025 rows with at least three prior series observations

C. Selective Prediction and Abstention

Selective classification and reject-option systems trade coverage for risk: a predictor may abstain when its risk target is not satisfied [26], [27]. Our unit is different. The decision is not a per-row class label but a study-level permission to interpret a fitted feature family. The protocol therefore couples a predictive-adequacy module with a paired feature-group value module and reports false permission and false abstention on synthetic scenarios with known data-generating mechanisms and pre-specified evaluation labels.

D. Evaluation and Reproducibility

Random splits can be inappropriate when temporal or hierarchical structure is present [28]. Leakage can create seemingly strong scientific machine-learning results while invalidating the intended forecast claim [29]. Reproducible ML also requires explicit code, data paths, settings, and numerical evidence [30]. Accordingly, every manuscript number is generated from an auditable run: configuration, prediction vectors, bootstrap draws, citation check, and final PDF are all versioned outputs.

III. DATA AND PROTOCOL

A. Panel, Features, and Target

The unit is a crop–state–year row. The processed panel contains 1257 observations from 1990–2025 for Barley, Canola, Oats, and Wheat across 12 U.S. states: Colorado, Illinois, Iowa, Kansas, Minnesota, Montana, Nebraska, North Dakota, Oklahoma, South Dakota, Texas, and Washington. The observed panel has 40 crop–state series, with available-year counts ranging from 10 to 36 (median 36). Yield in t ha^{-1} comes from USDA NASS Quick Stats [31]; daily maximum/minimum temperature, precipitation, and shortwave radiation come from NASA POWER [32]. The primary design has 35 full-season weather features. A separate sensitivity design adds early-, middle-, and late-season proxies without redefining the target or locked population. Because the primary features require a completed crop-season weather window, the task is a post-season scientific audit of residual-model reliance, not a pre-harvest forecast. Supplementary Table S2 defines each weather feature by model name, NASA POWER field, aggregation, window, unit, spatial rule, missing handling, and availability.

For a test interval, each crop–state linear trend and residual scale is estimated using only its associated training rows. A row is eligible only with at least three training observations. The validation period begins with 144 candidate rows and retains 140 after 4 pre-specified exclusions. The locked test

starts with 343 candidates, excludes 10, and evaluates 333 rows. Future yields cannot change a previously constructed evaluation target; this property has a unit test and row-level audit. The learned target is the raw train-only residual in t ha^{-1} : $\hat{y}_{c,s,t} = a_{c,s}^{(\text{train})} + b_{c,s}^{(\text{train})}t$, $r_{c,s,t} = y_{c,s,t} - \hat{y}_{c,s,t}$, and $z_{c,s,t} = r_{c,s,t} / \max(\hat{\sigma}_{c,s,\text{train}}, \epsilon)$. The standardized residual defines events, $\mathbb{K}[z_{c,s,t} < \tau]$, only. This distinction prevents a pooled raw-error score from being misread as a standardized event-recovery metric. Min-history rules of 3, 5, 8, and 10 years are reported as sensitivity analyses because both eligibility and the fitted residual scale can change with available history. Three observations are the minimum that leaves one positive residual degree of freedom for estimating a training residual scale; this is not claimed to provide a stable series-specific variance estimate. The model matrices exclude year, yield, trend, residual, standardized residual, event labels, predictions, history length, and residual scale; the no-shortcut schema and overlap audit are released with the artifact.

B. Selection and Baselines

Ridge $\alpha \in \{1, 10\}$ and Random Forest/ExtraTrees with 160 trees and $\text{min_samples_leaf} \in \{1, 2\}$ are evaluated on metadata-only, weather-only, and full representations. Stochastic fits use fixed seeds $\{7, 17, 27, 37, 47\}$; numeric imputation/scaling, categorical encoding, and model fitting occur within the training fold. Mean per-row seed-aggregated validation RMSE chooses the configuration, with lexicographic tie-breaking and no locked-test access. Seed-level RMSE standard deviations are reported so tiny differences are not presented as deterministic wins.

The final audit compares learned predictions on identical rows to zero residual, training-mean residual, and crop training-mean residual baselines. Prediction vectors are hashed and pairwise maximum absolute differences are retained, so any nearly coincident baselines are demonstrated rather than assumed. We use 2,000 deterministic calendar-year block bootstrap replicates and percentile 95% intervals over the ten locked calendar-year blocks. This preserves cross-sectional dependence within a year but limits interval precision. All 2,000 bootstrap replicates were valid and retained. We do not implement the stationary bootstrap itself; we cite it for the dependent-resampling motivation and resample complete calendar years as paired blocks [33]. State-year and crop–state cluster schemes are reported as pre-specified sensitivity analyses.

C. Pre-specified Claim Modules

Let P^* denote the locked target population implied by the split, target construction, feature set, and evaluation period. Module A estimates $\theta_A = \text{RMSE}_{P^*}(f^*) - \text{RMSE}_{P^*}(f_0)$, where f^* is the validation-selected model and f_0 is the zero-residual baseline. Module B estimates $\theta_B = \text{RMSE}_{P^*}(f_{M+W}) - \text{RMSE}_{P^*}(f_M)$, the incremental weather value beyond metadata. The reported estimates replace P^* by the locked empirical rows, use paired calendar-year block

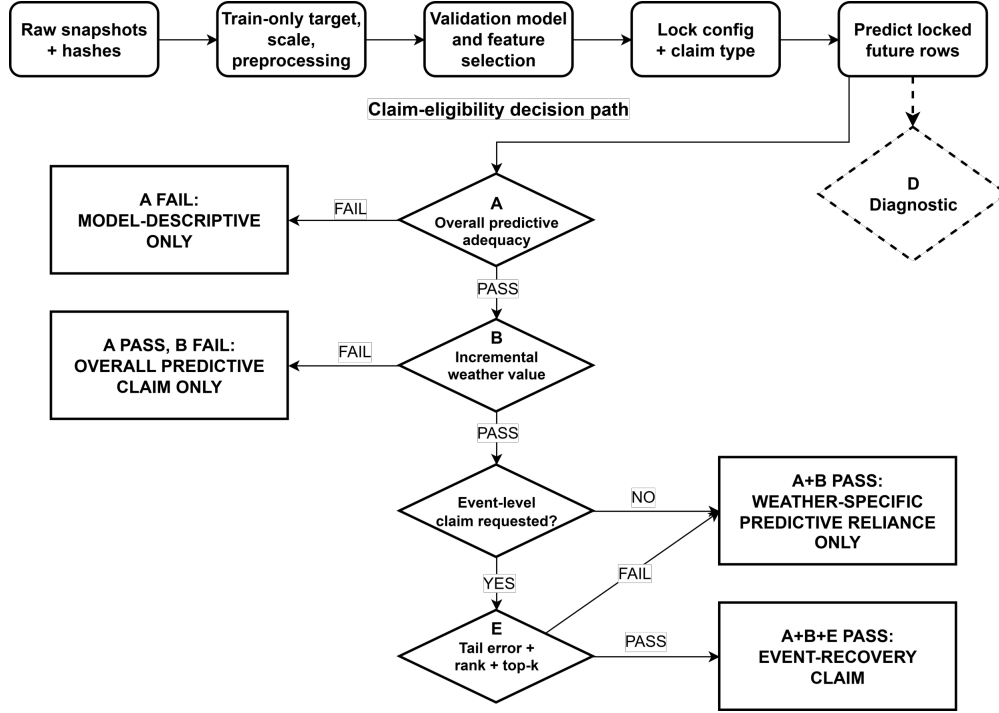


Fig. 1. Sequential claim-eligibility workflow. Module A determines whether claims may extend beyond model description, Module B controls weather-specific predictive reliance, and Module E is evaluated only for an event-level claim. Failure of Module E blocks event-recovery interpretation but retains the weather-specific predictive-reliance claim when Modules A and B pass. Module D is a diagnostic outside the permission path.

resampling, and pass only when the upper 95% interval is below zero.

Module E is invoked only for below-trend event claims. Its primary tail is $z < -1$; $z < -1.5$ and $z < -2$ are sensitivities because their smaller samples reduce power. The event module requires tail RMSE/MAE improvement, rank recovery with a positive lower bootstrap bound and within-year permutation $p < .05$, and top- k recovery above chance by lift, hypergeometric probability, permutation p , and year-block lift interval. Module D is the Weather-only versus Metadata-only representation diagnostic and is recorded outside the permission path. The fixed policy is in Table II.

Algorithm 1: Locked claim-eligibility audit

Input: split, candidate grid, feature groups, baseline, requested claim type.

1. Fit target construction and preprocessing on training rows only.
2. Select model/configuration on validation; lock model, feature roles, and claim rule.
3. Refit through the validation cutoff and predict locked future rows.
4. Estimate Module A on paired locked rows.
5. Estimate Module B with Full and Metadata-only representatives.
6. If an event-level claim is requested, estimate Module E on the primary tail.
7. Record Module D as a diagnostic outside the permission path.
8. Return the highest claim level whose required modules pass.

IV. RESULTS

A. Validation and Locked Test

Table III shows leading validation configurations. The selected model is ExtraTrees with weather only features. The complete grid, seed-level metrics, tie rule, and locked-test access flag are supplied in the artifact manifest. Module A uses this single overall validation-selected configuration.

Module B is a pre-specified fixed-architecture contrast: Full and Metadata-only predictions reuse the Module A configuration ID and swap only the feature family. Their metrics and row/target/prediction hashes are recorded in the selection manifest; they are not candidates for replacing Module A after the locked period is observed.

The selected weather-only ExtraTrees configuration ranks first in 3/5 stochastic validation seeds. Its advantage is small relative to several seed standard deviations, so we retain the pre-specified selection without claiming clear superiority. The one-standard-error flag and every seed-level score are retained; no final-test statistic changes this selection.

On the locked test, the selected model has $R^2 = -0.014$ and RMSE 0.669 t ha^{-1} . Its paired RMSE difference from zero crosses zero, so Module A fails. The selected configuration deteriorates from validation RMSE 0.384 in 2012–2015 to locked-test RMSE 0.669 in 2016–2025, consistent with the temporal instability summarized below. Table IV includes all same-task naive baselines and the selected configuration across feature families. Module B compares Full versus Metadata-only on the same locked rows. Module D compares Weather-only and Metadata-only as an exploratory representation diagnostic outside the permission path. All comparisons use paired predictions on identical locked rows. Modules A and B fail; Module D shows no clear difference and remains diagnostic.

The Metadata-only vector nearly coincides with the zero-residual baselines on the locked rows. The target is an OLS residual constructed separately within each crop-state

TABLE II

FIXED CLAIM MODULES. MODULE A CONTROLS OVERALL PREDICTIVE CLAIMS, MODULE B CONTROLS WEATHER-SPECIFIC PREDICTIVE RELIANCE, MODULE E CONTROLS BELOW-TREND EVENT-RECOVERY CLAIMS, AND MODULE D IS DIAGNOSTIC ONLY. FULL CONFIGURATION DETAILS ARE IN THE REPRODUCIBILITY MANIFEST.

Module	Question	Pass condition	Claim scope
A – Overall adequacy	Does the locked model improve on the same-target baseline?	Upper paired 95% CI of $\Delta\text{RMSE} < 0$.	Overall predictive claim
B – Weather value	Does Full improve on Metadata-only on the same locked rows?	Upper paired 95% CI of $\Delta\text{RMSE} < 0$.	Weather-specific predictive reliance
E – Event recovery	Does the model recover the primary below-trend tail beyond null checks?	Tail RMSE/MAE, rank, permutation, and top- k checks all pass.	Observed below-trend event claim
D – Diagnostic	How does Weather-only compare with Metadata-only?	Reported only; not a decision module.	Descriptive diagnostic only

TABLE III

VALIDATION SELECTION ON 2012–2015. RMSE IS COMPUTED AFTER FIXED-SEED PREDICTION AGGREGATION; SEED SD IS THE ACROSS-SEED RMSE STANDARD DEVIATION, AND $P(\text{RANK } 1)$ IS THE SEED-LEVEL FIRST-RANK FREQUENCY FOR STOCHASTIC CANDIDATES; RIDGE IS DETERMINISTIC AND SHOWN AS N/A. ROLE SEPARATES THE MODULE A WINNER FROM THE FIXED-ARCHITECTURE MODULE B/D REPRESENTATIVES.

Model	Features	Role	n	RMSE	Seed SD	$P(\text{rank } 1)$	1-SE	Selected
ExtraTrees (leaf=1)	Weather only	Module A: D	140	0.384	0.003	0.60	Yes	Yes
ExtraTrees (leaf=2)	Weather only	Validation candidate	140	0.385	0.001	0.20	No	No
Random Forest (leaf=1)	Weather only	Validation candidate	140	0.387	0.003	0.20	No	No
Random Forest (leaf=2)	Weather only	Validation candidate	140	0.387	0.002	0.00	No	No
ExtraTrees (leaf=2)	Full	Validation candidate	140	0.389	0.003	0.00	No	No
Random Forest (leaf=1)	Full	Validation candidate	140	0.390	0.002	0.00	No	No
ExtraTrees (leaf=1)	Full	Module B Full	140	0.391	0.003	0.00	No	No
Ridge ($\alpha = 10$)	Metadata only	Best Ridge candidate	140	0.407	N/A	N/A	No	No
ExtraTrees (leaf=1)	Metadata only	Module B/D Metadata	140	0.407	0.000	0.00	No	No

series, so its training mean is approximately zero within each fitted series. Metadata features are time-invariant context variables (latitude, longitude, region, and crop) and provide no within-series temporal signal. In the locked ExtraTrees configuration, the maximum absolute prediction difference is 1.32×10^{-13} versus the zero-residual vector, 1.13×10^{-13} versus the training-mean baseline, and 1.05×10^{-13} versus the crop-training-mean baseline. Consequently, Module D is numerically identical to Module A in this panel, although the contrasts answer different conceptual questions. Module B remains distinct because the Full model can use weather jointly with metadata.

B. Event-Detection Diagnostics

Event detection is evaluated over all 333 locked rows, not only a tail preselected by the observed outcome. The score is negative predicted residual and the score threshold is fixed at zero before final evaluation. Event scores retain weak global discrimination: ROC-AUC is 0.641 for the primary $z < -1$ tail and 0.672 for the two severe-tail sensitivities; primary-tail PR-AUC is 0.338 versus prevalence 0.219. This does not contradict the module failure. ROC-AUC summarizes pairwise ordering over all locked rows, whereas Module E asks whether the model improves paired error and recovers the pre-specified tail beyond rank and top- k null checks. The available discrimination is diagnostically positive but insufficient for the requested observed-event claim.

C. Below-Trend Event Module and Robustness

The broad below-trend subset contains 73 locked-test rows. Table V separates error, rank-null, and chance-adjusted event-recovery components. The primary $z < -1$ top-10 lift is below random expectation and its permutation result is not significant; rank uncertainty also crosses zero. Module E therefore fails without relying on a directional-sign statistic. Because Modules A and B fail, neither the local nor global explanation outputs can be used to support observed-event or weather-specific predictive claims in the primary analysis.

The locked Weather-only model still produces coherent fitted-function descriptions: grouped SHAP mass is concentrated in weather features, and the local decomposition in Fig. 2 assigns one below-trend row to recognizable weather groups. These outputs describe how the fitted predictor varies with its inputs. They are therefore reported as model diagnostics rather than evidence about observed yield losses.

The robustness artifact gives exact temporal pass counts and summary ranges for the pre-specified stage-feature, crop, spatial, and target-protocol checks. These checks do not reverse the central decision; they show its stability boundary rather than optimize around it.

D. Target, Cluster, and Model-Family Sensitivity

The three-observation rule is the pre-specified technical minimum, not an empirically optimal or stable variance threshold. Eligibility is audited at each validation/final cutoff. Histories 8 and 10 retain 313 rather than 333 final rows, so their favorable intervals are sensitivity results from a different population and do not alter the primary Module A decision. The audited mechanism is structural: Canola–Oklahoma has seven pre-2016 observations and Canola–Washington has five, so both series are removed by the eight- and ten-year filters; all retained series have at least ten pre-2016 observations. Thus the History 8 and 10 eligible row IDs and target vectors are identical; their predictions differ only at floating-point precision (maximum absolute difference 1.11×10^{-16}), so the displayed metrics coincide. The artifacts separate target/model checks from resampling, so row counts and cluster counts cannot be confused. The retained sensitivity rows are History 3, History 5, History 8, History 10, Standardized target, Huber detrending, HistGradientBoosting, and ElasticNet.

TABLE IV

LOCKED 2016–2025 SAME-TASK AUDIT. LEARNED-MODEL PREDICTIONS ARE SEED AGGREGATED. ΔRMSE IS MODEL MINUS ZERO-RESIDUAL RMSE WITH PAIRED 95% YEAR-BLOCK INTERVALS; RMSE IS THA^{-1} .

Model	Features	n	Seeds	R^2	RMSE	ΔRMSE vs zero [95% CI]
Crop train mean	Baseline	333	1	-0.031	0.674	0.000 [0.000, 0.000]
Train mean	Baseline	333	1	-0.031	0.674	0.000 [0.000, 0.000]
Zero residual	Baseline	333	1	-0.031	0.674	0.000
ExtraTrees	Full	333	5	0.006	0.662	-0.012 [-0.029, 0.002]
ExtraTrees	Metadata only	333	5	-0.031	0.674	0.000 [0.000, 0.000]
ExtraTrees	Weather only	333	5	-0.014	0.669	-0.005 [-0.019, 0.009]

TABLE V

TAIL COMPONENTS ON TRAIN-ONLY TARGETS. PANEL A REPORTS ERROR AND RANK RECOVERY; PANEL B REPORTS CHANCE-ADJUSTED TOP-10 RECOVERY. H IS THE HYPERGEOMETRIC PROBABILITY AND P IS THE WITHIN-YEAR PERMUTATION PROBABILITY. THE TOP-10 COMPONENT PASSES ONLY WHEN $\text{LIFT} > 1$, THE LOWER $\text{LIFT CI} > 1$, AND BOTH H AND P ARE $< .05$. ONLY $z < -1$ IS PRIMARY; $z < -1.5$ PASSES ERROR BUT FAILS RANK AND TOP-K, WHILE $z < -2$ FAILS THE COMPLETE EVENT MODULE.

Panel A: Error and rank recovery

Role	Tail	n	ΔRMSE [95% CI]	ΔMAE [95% CI]	Rank ρ [95% CI]	Perm. p	Status
Primary	$z < -1$	73	-0.043 [-0.074, 0.007]	-0.054 [-0.096, 0.002]	0.180 [-0.035, 0.373]	0.362	FAIL
Sensitivity	$z < -1.5$	52	-0.056 [-0.083, -0.002]	-0.073 [-0.109, -0.007]	0.060 [-0.151, 0.416]	0.511	FAIL
Sensitivity	$z < -2$	36	-0.056 [-0.090, 0.014]	-0.076 [-0.118, 0.015]	-0.051 [-0.306, 0.497]	0.797	FAIL

Panel B: Chance-adjusted Top-10 recovery

Role	Tail	n	Observed	Expected	Lift [95% CI]	H p	P p	Status
Primary	$z < -1$	73	1/10	1.37	0.73 [0.00, 2.82]	0.794	0.793	FAIL
Sensitivity	$z < -1.5$	52	1/10	1.92	0.52 [0.37, 2.19]	0.907	0.935	FAIL
Sensitivity	$z < -2$	36	1/10	2.78	0.36 [0.32, 1.68]	0.979	0.985	FAIL

TABLE VI

LOCKED MODULE DECISIONS. MODULE A IS REQUIRED FOR OVERALL PREDICTIVE CLAIMS, MODULE B FOR WEATHER-SPECIFIC PREDICTIVE RELIANCE, MODULE E FOR BELOW-TREND EVENT-RECOVERY CLAIMS, AND MODULE D IS DIAGNOSTIC ONLY.

Mod.	Role	Contrast/check	Estimate or primary statistic	Status	Decision role
A	Overall predictive adequacy	Selected Weather-only vs. zero residual	-0.005 [-0.019, 0.009]	FAIL	Blocks overall and higher claim levels
B	Incremental weather value	Full vs. Metadata-only	-0.012 [-0.029, 0.002]	FAIL	Blocks weather-specific predictive reliance
E	Event-recovery adequacy	$z < -1$ tail error, rank, and top-10 null checks	RMSE -0.043 [-0.074, 0.007]; rank 0.180; top-10 1/10	FAIL	Error CI, rank, and top-10 checks do not all pass
D	Representation diagnostic	Weather-only vs. Metadata-only	-0.005 [-0.019, 0.009]	DESCRIPTIVE ONLY	CI crosses zero; diagnostic only

Full target/model and paired-resampling tables are retained as supplementary tables; paths and hashes are listed in the reproducibility manifest.

The standardized-target model is actually refit on the train-only residual z , rather than obtained by rescaling raw-target predictions after the fact. It also changes the point-estimate direction: the raw-residual model is slightly better than zero at the point estimate ($\Delta\text{RMSE} = -0.005$), whereas the standardized-target model is slightly worse ($\Delta\text{RMSE} = 0.030$). Its paired interval still crosses zero. Test-period z values need not have unit variance: each crop-state series uses its training-only residual scale, and a small training scale can yield larger future standardized residuals without leakage.

Replacing ordinary least-squares detrending by a training-only Huber trend changes the primary event count from 73 to 40, but its paired interval still crosses zero. These analyses are deliberately reported as target-construction sensitivities, not candidate routes to a favorable result.

E. Temporal Stability and Capacity

The four rolling-origin folds use non-overlapping future blocks after 2003, 2006, 2009, and 2012 training cutoffs. Three of four folds have negative point estimates, but only the 2013–2015 fold has a 95% CI fully below zero. In contrast, the 2007–2009 fold significantly favors the zero-residual baseline ($\Delta\text{RMSE} = 0.017$, 95% CI [0.003, 0.028]). A prefix learning curve retrains the same validation-locked configuration using

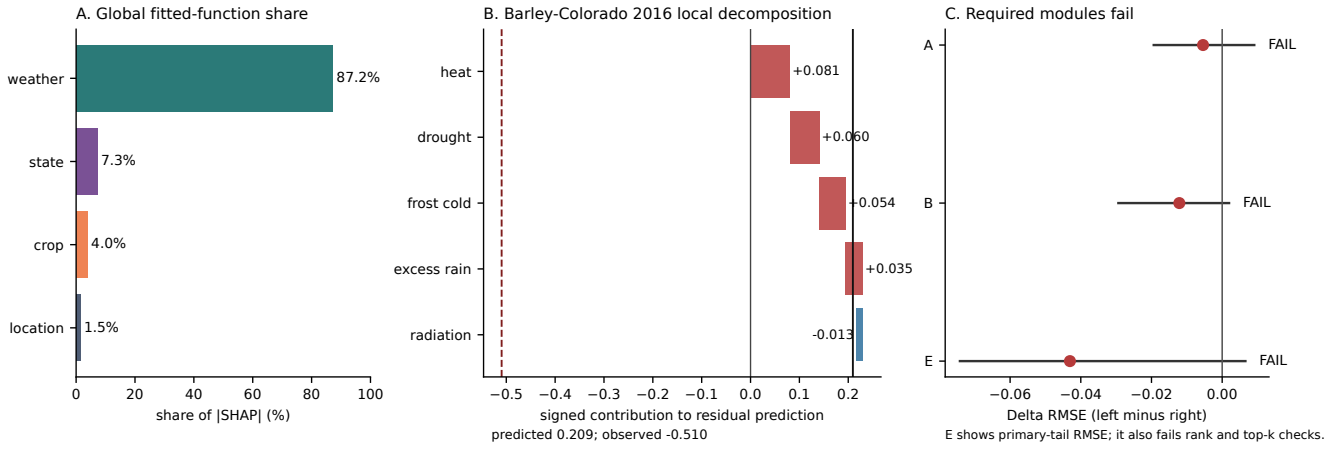


Fig. 2. Agricultural explanation output under abstention. SHAP family shares and the local signed group decomposition describe the locked fitted Weather-only model, while Modules A, B, and E fail on locked outcomes. These are fitted-function diagnostics, not event-recovery evidence.

progressively longer historical prefixes, while retaining only crop–state series with at least three prefix observations. None of the four prefix intervals establishes improvement over zero. This makes the conclusion conditional on the available panel and model family, not a claim that more data cannot help.

The exact rolling-origin and prefix learning-curve table is retained as a supplementary table with paths and hashes in the reproducibility manifest.

F. Audit Traceability and Interpretation Limits

The E1–E10 manifest maps each numerical claim family to a concrete artifact: E1–E4 cover splits, train-only targets, validation, resampling, predictions, and tail metrics; E5–E10 cover feature timing, crop/spatial checks, target construction, baselines, and tables. Full paths, commands, and hashes are in the released acceptance audit and E1–E10 manifest.

Interpretation limits are summarized in Discussion; none justifies relaxing the locked rule after evaluation.

G. Retrospective Target Sensitivity

Full-series detrending uses later yields to construct earlier residuals. It is therefore a retrospective diagnostic, not a prospective target. Relative to the train-only construction, the $z < -1$ final-period set has Jaccard overlap 0.605 with year-block 95% interval [0.500, 0.675]; 21 prospective events are not retrospective events and 13 retrospective events are not prospective events. A coherent retrospective explanation would still not repair the prospective fidelity failure.

The rendered retrospective comparison is retained in the supplementary artifact set and indexed by the reproducibility manifest.

V. CALIBRATION AND EXTERNAL CASES

A. Synthetic Benchmark

The claim-module policy is also evaluated where the mechanism and claim label are known before evaluation. A repeated synthetic temporal benchmark covers 14 regimes over 30

State-level locked Delta RMSE: Weather-only minus zero residual

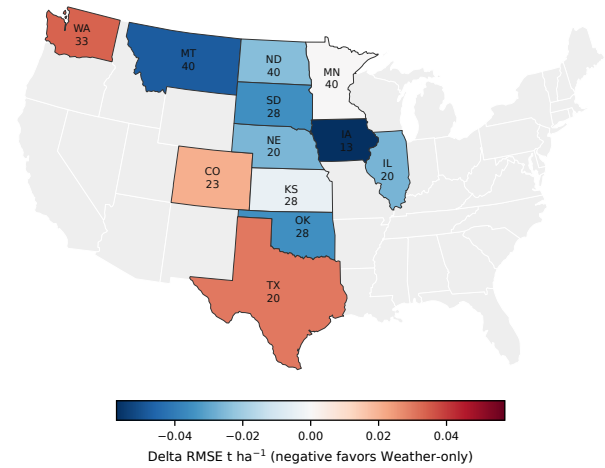


Fig. 3. State-level heterogeneity on the locked 2016–2025 crop panel. Colors show the descriptive difference in RMSE between the validation-selected Weather-only model and the zero-residual baseline within each state; negative values favor the model. Labels report the state abbreviation and the number of locked test rows. Gray states are outside the study panel. These subgroup estimates illustrate spatial heterogeneity and are not separate module decisions. State geometry is obtained from the U.S. Census Bureau [34]; processing settings and hashes are recorded in the map manifest.

seeds each, including correlated features, omitted confounding, temporal and geographic shift, and leakage. GT permit labels define six permissible regimes and are used only for scoring; policy permit is computed from observable Module A, B, and E outputs. Ungated explanations falsely permit 240 of 240 invalid runs (100.0%, 95% CI [98.4,100.0]); the observable A+B+E policy falsely permits 171/240 invalid runs (71.2%, 95% CI [65.2,76.6]) while falsely abstaining on 20/180 valid runs (11.1%, 95% CI [7.3,16.5]). Its permission rate is 78.8%, sensitivity 88.9%, and specificity 28.7%.

Synthetic scenario-level calibration. GT permit is used only as the evaluation label. Policy permit is computed solely from pre-specified modules and observable checks; the GT label and hidden data-generating mechanism are not inputs to the policy. A, B,

E, and Policy report pass or permission rates across 30 seeds for an event-recovery claim.

Scenario	GT	Claim	A	B	E	Policy
Corr. feat.	no	event	100%	100%	100%	100%
Geo. shift	no	event	100%	100%	100%	100%
Imbalanced tail	yes	event	100%	100%	100%	100%
Leakage	no	event	100%	100%	100%	100%
Meas. error	no	event	70%	100%	83%	67%
Moderate signal	yes	event	100%	100%	100%	100%
No signal	no	event	0%	47%	10%	0%
Omitted conf.	no	event	100%	100%	100%	100%
Small sample	yes	event	93%	97%	97%	93%
Spatial mismatch	yes	event	43%	100%	80%	40%
Strong signal	yes	event	100%	100%	100%	100%
Temporal drift	no	event	97%	100%	100%	97%
Train-only detr.	yes	event	100%	100%	100%	100%
Weak signal	no	event	10%	80%	53%	7%

The observable modules cannot reliably identify all structurally invalid regimes: omitted confounding, geographic shift, leakage, and correlated features are permitted in 30/30 seeds, and temporal drift in 29/30. These are false permissions, not successful abstentions.

B. Agricultural External-Resolution Check

We also implemented a separate county-level winter-wheat check rather than assuming that a finer panel would rescue the state-level result. The locked pre-model population contains 574 counties with at least ten training-period observations from 2000–2025, official county yields, static county interior points, and annual NASA POWER weather aggregates. Validation selected a weather-only ExtraTrees residual model from Ridge and ExtraTrees metadata, weather, and full-feature candidates. On the untouched 2022–2025 holdout, its RMSE was 13.47 versus 13.78 bushels per acre for the zero-residual baseline, but the paired year-block 95% CI for the difference was $[-0.67, 0.26]$. Module B, which compares Full with Metadata-only, passes ($\Delta\text{RMSE} = -0.71$, 95% CI $[-1.23, -0.34]$), but Module A remains inconclusive. The county analysis therefore reaches a model-descriptive-only outcome and serves as a robustness case rather than evidence of predictive success or cross-resolution transfer.

C. Cross-Domain Sanity-Check Permission Case

The PJM analysis is a cross-domain sanity-check permission case for the core two-stage policy rather than a transfer claim. Its target is continuous electricity demand, so the crop-specific extreme-event module is not invoked. Public daily PJM electricity-demand and day-ahead forecast records from EIA Form EIA-930 [35] are fit with Calendar-only ExtraTrees features on January–September 2024 and evaluated on locked October–December records; Full adds the day-ahead demand forecast. Module A compares Full with a pre-specified train-period mean-demand naive baseline and passes: locked RMSE is 77,521 versus 335,037 MWh, with paired bootstrap ΔRMSE 95% CI $[-296.7, -221.8] \times 10^3$ MWh. Module B compares Full with Calendar-only and also passes: Calendar-only RMSE is 227,417 MWh and the Full-minus-Calendar paired 95% CI is $[-186.0, -119.3] \times 10^3$ MWh. Because both pass, permutation importance is released for locked PJM rows. The XAI manifest binds row IDs, hashes, seed handling, and output

scale; agricultural XAI artifacts remain descriptive because the required agricultural modules fail. These are predictive-reliance interpretations, not causal or transfer claims. This intentionally easy positive case checks that the policy is not always-abstaining; synthetic regimes define the policy boundary.

VI. DISCUSSION

A. Claim Scope and Multiplicity

Several analyses are locally favorable. The eight- and ten-year history filters improve paired error on a smaller 313-row population; the 2013–2015 rolling fold passes, whereas the 2007–2009 fold significantly favors the baseline; severe tails improve RMSE and MAE but fail rank and top- k recovery; and the county analysis passes Module B while Module A remains inconclusive. These results identify design conditions under which predictive signal may be stronger, but they either change the population, represent non-primary temporal folds, or fail another required module. We therefore use them to motivate future study design rather than to replace the primary locked test.

The audit separates primary decisions from descriptive measurements before the locked period is examined. Module A, primary Module B, and the primary $z < -1$ Module E define the below-trend event permission rule; more severe tails, alternative histories, detrending choices, resampling units, stage proxies, crop/state diagnostics, and expanded model families are sensitivity or diagnostic evidence. They clarify robustness and design trade-offs but are not extra chances to declare the primary claim successful.

B. Reproducibility and Traceability

The release records input hashes, target construction, feature availability, forbidden-column checks, locked configuration, seeds, row IDs, predictions, bootstrap draws, and table inputs. Regenerated records are matched to the audit record, and each paired comparison can be recomputed on identical locked rows.

C. Limitations and External Validity

State-representative weather can miss within-state exposure, weather covariates are correlated, and the panel lacks soil, irrigation, cultivar, management, and phenology data. The primary interval resamples only ten calendar-year blocks, and cluster sensitivities encode different dependence assumptions without removing that limitation. This negative result limits the present target, resolution, features, and model family; it does not establish that weather lacks scientific relevance for yield. The corrected synthetic benchmark also shows that predictive adequacy and feature-group value cannot identify omitted confounding, leakage, correlated-feature structures, or unobserved distributional violations by themselves. GT labels are used only for evaluation, not for policy decisions. Future work needs prospectively versioned inputs, finer exposure measurements, design-stage validity checks, and an independent future test interval.

VII. CONCLUSION

This paper evaluates whether an interpretable weather-only model can support claims about below-trend U.S. crop-yield events after validation-locked selection. Modules A, B, and E fail for the state panel, and a separately locked county-level winter-wheat panel remains inconclusive under Module A. Thus this fitted model may be described, but it does not support observed-event or weather-specific predictive-reliance claims in this panel. More generally, locked targets, same-row baselines, paired uncertainty, and module separation define the highest supported claim level. The corrected synthetic benchmark's 88.9% sensitivity and 28.7% specificity show that the modules are necessary for claim discipline but not sufficient to detect structural invalidity; leakage, omitted confounding, measurement validity, and distribution-shift checks remain design-stage requirements.

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